

Vineeth Raghav Gururajan

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PROFILE

M.Sc. Electrical Engineering student · Power Electronics · SPICE Circuit Simulation (PSpice) · Circuit Analysis & Validation

- 1 year of industry experience in circuit design and validation, component evaluation and selection, and test automation in Python
- Coursework in power electronics, battery modeling and power systems, aimed at onboard chargers and DC/DC converters in electric mobility; PSpice certified
- Available as a working student up to 20 h/week, flexible start; based in Stuttgart

EXPERIENCE

Electronic Design Intern – FocalSystech Pvt. Ltd., Chennai, India 07/2024 – 07/2025

- **Circuit design & validation:** Designed and validated electronic circuits for embedded systems; verified designs against technical requirements through simulation and measurement on prototypes
- **Component evaluation & selection:** Evaluated and selected components against technical requirements, working from schematics and datasheets
- **Python automation:** Automated testing and performance analysis in embedded C and Python, reducing manual test effort
- **PCB & integration:** Created PCB layouts in Altium, supported prototype fabrication and bring-up, and defined interfaces between sensors, actuators and microcontrollers (CAN, I2C, SPI, UART)

EDUCATION

M.Sc. Electrical Engineering – University of Stuttgart 10/2025 – expected 2027

- Intelligent Power Grids, Power Systems Engineering, Battery Modeling, Power Electronics, Control Systems

B.Tech. Electronics and Computer Engineering – SRM Institute of Science and Technology, Chennai · GPA 8.7/10 2021 – 2025

PROJECTS

Battery Cell Modeling – Team project, University of Stuttgart 2026

- Analysed cell-to-cell variance caused by electrical, thermal, aging and manufacturing tolerances by simulation; implemented and debugged statistical parameter sampling in the model. Directly relevant to battery pack and battery management design

64-bit RoBA Multiplier with MAC Unit – B.Tech. final-year project, Verilog 2025

- Replaced the Kogge–Stone adder with an area- and power-efficient Han–Carlson adder; synthesised and validated the design for improved speed, power and area. An optimisation exercise: same function, lower cost and complexity

SKILLS

Simulation & Analysis	SPICE circuit simulation (PSpice), circuit design and validation, statistical tolerance analysis in simulation (battery project)
Power & E-Mobility	Power electronics, battery modeling, control systems; PCB design (Altium), component evaluation
Software & Tools	Python (automation), MATLAB, C/C++, Verilog; microcontrollers (ARM, ESP32, Arduino)
Integration & Test	CAN, I2C, SPI, UART, system integration, test automation, debugging

LANGUAGES & OTHER

English: fluent · **German:** A2, preparing for the telc B1 exam

Certificates: PSpice Fundamentals (TVS India), Certificate of Training (Livewire Ltd.) · **Volunteering:** STEM workshops for underprivileged students; post-cyclone relief in Chennai